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Joint Variable Selection in Generalized Linear Mixed Models With Random Regularized Penalized Quasi-Likelihood Technique

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Abstract
Variable selection provides a pathway toward preventing generalized linear mixed models from encountering issues of model overfitting, nonconvergence, low external validity, and estimation biases. Among the various selection approaches, the methods of regularized penalized quasi-likelihood (rPQL) show some superiority in jointly choosing both important fixed and random effects. However, three challenges limit the wider usage of the variable selection approaches in practice: the high computational cost, the outnumbered predictors problem, and multicollinearity. To overcome these challenges, the current study proposes a new algorithm, the random rPQL, that incorporates an rPQL estimation with the resampling technique. In addition, the study introduces a new selection criterion, the ranking worth estimation, to the selection process. Simulation results indicate (a) random rPQL can select fixed and random effects with high accuracy and efficiency even when the number of candidate variables exceeds within-group observations or when severe multicollinearity exists. (b) The results also show that the ranking worth estimation is more robust in the progress of regularization integrated with the resampling approach.

Translational Abstract
Variable selection provides a pathway toward preventing generalized linear mixed models from encountering issues of model overfitting, nonconvergence, low external validity, and estimation biases. Among the various selection approaches, the methods of regularized penalized quasi-likelihood (rPQL) show some superiority in jointly choosing both important fixed and random effects. However, three challenges limit the wider usage of the variable selection approaches in practice: the high computational cost, the outnumbered predictors problem, and multicollinearity. To overcome these challenges, the current study proposes a new algorithm, named random rPQL, that incorporates an rPQL estimation with the resampling technique. In addition, the study introduces a new selection criterion, the ranking worth estimation, to the selection process. Simulation results indicate random rPQL can select fixed and random effects with high accuracy and efficiency. The proposed ranking worth estimation is more robust in the progress of regularization integrated with the resampling approach.

Keywords: variable selection, generalized linear mixed models, regularization, high dimensionality

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One major task in social and behavioral research is to describe, explain, and predict human behaviors by building statistical models with empirical data (Breiman, 2001; Yarkoni & Westfall, 2017). However, this task becomes significantly challenging when the data contain limited observations (n) but a large number of predictors (p). If building a model involves many available predictors, it will capture the noise and idiosyncratic sample-specific characteristics rather than generalizable patterns—an issue commonly referred to as model overfitting (Dwyer et al., 2018). In this situation, the generalizability becomes a legitimate concern—models built in one sample may not be applicable to other samples drawn from the same population. In addition, when $p > n$, model coefficients cannot

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be uniquely estimated, and computational failures occur (Johnstone & Titterington, 2009). To overcome these challenges, variable selection has been highly recommended prior to any null hypothesis testing (Jonas & Cesario, 2016; Yarkoni & Westfall, 2017). It refers to the techniques aided in selecting important predictors while removing nonimportant ones from all candidates so that the subsequent analyses can enhance computational efficiency, prevent model overfitting, and improve generalizability of research findings (Dwyer et al., 2018; McNeish, 2015; Van Der Maaten et al., 2009).

The past decades have witnessed a rapid development of variable selection techniques (e.g., Bozdogan, 1987; Claeskens & Hjort, 2008; Matuschek et al., 2017; Stroup, 2012; Vergouwe et al., 2010). For instance, in the context of generalized linear models (GLMs), a variety of selection benchmarks have been proposed, such as penalized likelihood (J. Fan & Li, 2001; Tibshirani, 1996; Zou & Hastie, 2005) and change-in-estimate criterion (Bursac et al., 2008; Maldonado & Greenland, 1993; Vansteelandt et al., 2012). In recent years, research on variable selection shifted its focus to generalized linear mixed models (GLMMs), a class of models widely applied to model clustered data with both fixed and random effects. Given that both types of effects need to be considered in a variable selection process, it is unsurprisingly challenging to select variables in the context of GLMMs. This study aims to propose an innovative method for joint variable selection in GLMMs that could overcome major limitations of the existing methods.

Among the approaches for variable selection, regularization is deemed to be effective and feasible. The regularization techniques (e.g., least absolute shrinkage and selection operator [LASSO], adaptive LASSO, and smoothly clipped absolute deviation) have demonstrated high accuracy in selecting important variables for GLMs (e.g., J. Fan & Li, 2001; Tibshirani, 1996; Zou & Hastie, 2005). The performance of these regularization techniques can be further improved by using resampling techniques (e.g., Liu et al., 2016; Wang et al., 2011). The main idea is to repeatedly apply a regularization procedure to bootstrapped samples and then decide the importance of all candidate variables according to their frequency of nonzero regression coefficients across all bootstrapped samples. Research has shown that the combination of regularization and resampling techniques can tackle issues of multicollinearity and thus enhance selection stability (Bach, 2008; Kim et al., 2019; Meinshausen & Bühlmann, 2006; Wang et al., 2011). So far, the implementation of regularization procedures in bootstrapped samples has been exclusive to GLMs, where only the fixed effects are estimated and involved in variable selection.

When regularization is used to select variables in GLMMs, one needs to perform the procedure on both fixed and random effects, which is also known as a joint selection (e.g., Bondell et al., 2010; Ibrahim et al., 2011). Researchers have proposed different regularization or shrinkage techniques to perform joint selection in GLMMs. For instance, Y. Fan and Li (2012) utilized penalized profile likelihood to select nonzero fixed effects and regularized posterior mode estimates to select nonzero random effects. B. Lin et al. (2013) improved this approach by using penalized restricted log likelihood to enhance estimation accuracy and robustness against outliers. Later, Pan and Shang (2018) optimized the penalized estimation by employing the penalized profile log likelihood on the fixed effects and the penalized restricted profile log likelihood on the random effects. Meanwhile, Hui et al. (2017) performed joint selection for both fixed and random effects via regularized penalized quasi-likelihood (rPQL) estimation, as discussed in a later section. Simulation results generally supported the accuracy of these methods for joint selection, albeit to a certain extent. However, several technical challenges remain.

First, joint selection is often associated with high computational costs. The absence of an analytic form for the log-likelihood function in GLMMs complicates the simultaneous regularization of both mean and variance structures. Therefore, most current methods use iterative processes, such as the expectation-maximization algorithm (Meng & Rubin, 1993) or the Newton–Raphson algorithm (Jennrich & Schluchter, 1986; Lindstrom & Bates, 1988), to approximate the intractable integral. However, the convergence of these processes can be extremely slow, which makes the computational burden to be unmanageable when the number of random effects gets large.

The second challenge arises when candidate predictors outnumber the available observations within a cluster. For example, in longitudinal studies, the number of prospective predictors, p, can exceed the number of repeated data collections, t. In such cases, methods for joint selection, such as those proposed by Bondell et al. (2010) and Ibrahim et al. (2011), are slow or even fail to converge. To address this issue, several methods have been proposed to select the fixed and random effects in separate steps (e.g., Y. Fan & Li, 2012; B. Lin et al., 2013; Pan & Shang, 2018). For example, Y. Fan and Li (2012) proposed replacing the unknown covariance of the random effects with proxy matrices in a modified likelihood function, thereby selecting the fixed and random effects in separate stages. This method could address issues when $p > t$; however, the dimensions of the fixed effects first need to be reduced while ignoring the random effects (Chen et al., 2014), which in turn could bias the estimates for the fixed effects (Pan & Shang, 2018).

Third, multicollinearity can be problematic when using regularized methods to select variables in GLMMs. Multicollinearity has been viewed as an ill-posed data issue that causes statistical problems, such as unstable parameter estimates, inflated standard errors, and unreliable inferences (Bingham & Fry, 2010; Kuhn & Johnson, 2013). While multicollinearity has been found not to bias the estimates for the fixed effects in GLMMs, it would magnify the standard errors of the estimates (Yu et al., 2015). Ideal regularization techniques can select a group of highly correlated predictors when they are all important to the outcome or deselect them simultaneously when they are all not important (Nguyen & Ng, 2020; Zou & Hastie, 2005). However, the popular LASSO penalty struggles with the issues of multicollinearity because it tends to arbitrarily select only one among all highly correlated, important variables (Meinshausen & Bühlmann, 2006; Wang et al., 2011; Zou & Hastie, 2005). Other regularized methods are similarly prone to randomly select one or more but not all highly correlated variables in GLMMs.

In short, the existing regularization methods are limited in dealing with the technical issues described above. The goal of this study is to propose a new method named random rPQL to perform a joint selection in GLMMs by using regularization techniques combined with the resampling procedure. For better application, we also developed an R package called random rPQL, which is currently accessible on the GitHub platform (https://github.com/var-selection/variable_selection). In the following sections, we first introduce a two-level GLMM and an existing regularized method, rPQL. Then, we propose the random rPQL in conjunction with a new selection criterion as an innovative expansion of rPQL. We then evaluate the performance of random rPQL through two simulation studies and illustrate its application via an empirical example. In the end, we discuss our findings and offer recommendations regarding the application of the proposed method.

GLMMs and rPQL

GLMMs are a class of models that estimate the relationship between outcomes and predictors while handling the dependency among individuals within the same clusters or time-ordered observations within the same individuals. This technique has been widely applied in various fields, such as health (Buka et al., 2003; Diez-Roux, 2000), psychology (Dearing et al., 2006; Long & Perkins, 2007; Rice et al., 1998), sociology (Lochner et al., 2001; Villemez & Bridges, 1988), and education (Maes & Lievens, 2003). GLMMs are built to estimate both fixed and random effects, with the former describing the relationship between the predictors and the outcome at the mean level and the latter estimating the variability of the coefficients across clusters or individuals.

A GLMM can be generically expressed as:

$$y_j = X_j\beta + Z_j u_j + \varepsilon_j, \tag{1}$$

or,

Equation 2 (see below)

Given a certain cluster j, the outcome variable y_j is a $n_j \times 1$ vector, and n_j is the total number of observations inside this jth cluster. The element of y_j vector is y_{ij}, devoting for the ith observations clustered in the jth cluster, where $i = 1, 2, \ldots, n_j$, $j = 1, 2, \ldots, m$, and m represents the total number of clusters. The total sample size, thereby, is $N = \sum_{j=1}^{m} n_j$. Furthermore, X_j, an $n_{\mathrm{j}} \times p$ matrix of the variables with fixed effects, of which the element x_{ijk} corresponds to the ith observation within the jth cluster for the kth variable, $k = 1, 2, \ldots, p$. Similarly, Z_j is a $n_{\mathrm{j}} \times q$ matrix of the variable's random effects. The element z_{ijk} is for the ith observation within the jth cluster for the lth variable, $l = 1, 2, \ldots, q$. The letters p and q denote the total number of fixed and random effects, respectively. The fixed effect β, a $p \times 1$ vector, is constant across all clusters. The random effect coefficients u_j, a $q \times 1$ vector, however, vary by clusters. It is assumed that $u_j \sim N(0, D)$, where D is a positive-definite covariance matrix. The residual ε_j, a $n_j \times 1$ vector, represents the within-cluster variability with $\varepsilon_j \sim N(0,\ \sigma_\varepsilon^2)$. The applications of GLMMs in psychology often take Z_j as a subset of X_j. For example, in longitudinal settings, the time variable is typically treated as both a fixed and a random effect, while other covariates are treated only as fixed effects. In fact, X_j and Z_j are allowed to differ. Any elements in Z_j but not in X_j, indicate their corresponding fixed effects coefficients β_j equal to 0.[1] For noncontinuous outcomes (e.g., binary, count), a link function $g(\cdot)$, such as the logit link, is used to relate the mean of the outcome to the linear predictor $X_j\beta + Z_j u_j$.

The fixed effects coefficients (β) and variance–covariance matrix (D) are the primary parameters to be estimated in GLMMs. However, the standard maximum likelihood estimation is not directly applicable to GLMMs due to a lack of an analytic form for the integral of the marginal log likelihood. Consequently, the parameter estimation for GLMMs necessitates computationally intensive iterative processes. To improve computational efficiency, the penalized quasi-likelihood (PQL; Breslow & Clayton, 1993; Dean & Nielsen, 2007) was recommended. The PQL estimator uses Laplace's approximation to approximate the integrated likelihood, thereby reducing computational time. Later, rPQL was proposed to conduct a joint selection with the PQL estimator (Hui et al., 2017).

The rPQL involves a two-step interactive procedure. Step 1 uses an adaptive LASSO (with a penalty parameter λ) to estimate the fixed effects coefficients ($\widehat{\beta_\lambda}$) across all clusters, conditioning on the random effects coefficients (U), and its corresponding variance–covariance matrix (D). Step 2 uses the adaptive group LASSO to update the random effects coefficients ($\widehat{U_\lambda}$), conditional on variance–covariance matrix (D), and fixed effects coefficients ($\widehat{\beta_\lambda}$) estimated from Step 1. The $\widehat{\beta_\lambda}$ and $\widehat{U_\lambda}$ are utilized with the Laplace approximation log likelihood to estimate variance–covariance matrix ($\widehat{D_\lambda}$). Through the regularization process, any coefficients associated with the nonimportant fixed and random effects are shrunk to zero. If a specific random effect coefficient shrinks to zero, this signifies the removal of its corresponding variance components from the variance–covariance matrix. The procedure iterates until convergence is achieved.

The performance of rPQL critically depends on the penalty parameter λ. The optimal penalty parameter (λ) is selected by using an information criterion (IC)[2] developed by Hui et al. (2017) . Specifically, for a given data set, a sequence of λ values is predetermined in a range of

[1] Typically, the applications of GLMMs in social and behavioral research, "fixed effects" denote the fixed effects coefficients β_j and "random effects" are the variance components in the variance–covariance matrix of the random effect coefficients (u_j) across clusters. In the literature of variable selection, the "random effects" and "fixed effects" refer to the predictors whose fixed effects coefficients or variance components are nonzero. The definition of fixed and random effects in the following sections aligns with the terminology used in the field of variable selection.

[2] $\mathrm{IC}(\lambda) = -\frac{2}{N} l_{PQL}(\widehat{\psi_\lambda}, \widehat{b_\lambda}) + \frac{\log(m)}{N}\dim(\widehat{\beta_\lambda}) + \frac{2}{N}\dim(\widehat{u_\lambda}),$

where $\dim(\widehat{\beta_\lambda})$ and $\dim(\widehat{u_\lambda})$ are the number of nonzero estimated fixed and random effects coefficients, respectively. N is the total sample size. $l_{\mathrm{PQL}}(\widehat{\psi_\lambda}, \widehat{b_\lambda})$ denotes the quasi-likelihood function. Be noted, $\mathrm{IC}(\lambda)$ is a combination of a BIC-type model complexity penalty for the fixed effects and an AIC-type penalty for the random effects.

$$\underbrace{\begin{bmatrix} y_{1j} \\ \cdots \\ y_{ij} \\ \cdots \\ y_{nj} \end{bmatrix}}_{\mathbf{y_j}} = \underbrace{\begin{bmatrix} 1 & x_{1j1} & \cdots & x_{1jp} \\ \cdots & \cdots & \cdots & \cdots \\ 1 & x_{ij1} & \cdots & x_{ijp} \\ \cdots & \cdots & \cdots & \cdots \\ 1 & x_{nj1} & \cdots & x_{njp} \end{bmatrix}}_{\mathbf{x_j}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \cdots \\ \beta_p \end{bmatrix}}_{\boldsymbol{\beta}} + \underbrace{\begin{bmatrix} 1 & z_{1j1} & \cdots & z_{1jq} \\ \cdots & \cdots & \cdots & \cdots \\ 1 & z_{ij1} & \cdots & z_{ijq} \\ \cdots & \cdots & \cdots & \cdots \\ 1 & z_{nj1} & \cdots & z_{njp} \end{bmatrix}}_{\mathbf{z_j}} \underbrace{\begin{bmatrix} u_{0j} \\ u_{1j} \\ u_{2j} \\ \cdots \\ u_{qj} \end{bmatrix}}_{\mathbf{u_j}} + \underbrace{\begin{bmatrix} \varepsilon_{1j} \\ \cdots \\ \varepsilon_{ij} \\ \cdots \\ \varepsilon_{nj} \end{bmatrix}}_{\boldsymbol{\varepsilon_j}}. \tag{2}$$

[$\lambda_{\min}$, $\lambda_{\max}$]— where $\lambda_{\min}$ corresponds to the full model containing all the candidate fixed and random effects, $\lambda_{\max}$ corresponds to the parsimonious model which only contains an intercept. Within this range, the optimal λ results in the minimum value of the IC.

The rPQL was shown to be superior to other regularized methods used for joint selection in GLMMs (Hui et al., 2017). However, it could fail to yield a reasonable solution when insufficient observations are present in a cluster, because the algorithm is not designed to handle the case where the number of random effects exceeds the number of observations within a cluster (X. Lin & Breslow, 1996). Additionally, rPQL utilizes an adapted LASSO technique, which makes rPQL inherit the LASSO's limitation in handling strong correlations among variables.

Random rPQL: An Innovative Expansion of rPQL for Joint Selection

We propose a random rPQL as a new joint selection method for GLMMs, aiming to overcome the limitations of the rPQL through utilizing resampling techniques. Before implementing random rPQL, researchers need to designate candidate predictors as either fixed, random, or both. The implementation of random rPQL requires two stages of tasks. Stage 1 starts from generating bootstrapped samples by randomly drawing observations with replacements from the original data. Within each bootstrapped sample, two subsets of predictors are randomly and respectively drawn from the corresponding candidate fixed and random effects. One subset will be used for the fixed effects, and the other one will be used for the random effects in a GLMM. Because of the random draws, predictors in the two subsets will not be identical. The subsequent implementation of rPQL in each bootstrapped sample yields the estimates for the fixed and random effects. An index for all candidate predictors is computed as the arithmetic mean of the absolute values of the estimates across all bootstrapped samples. This index is then used to determine each predictor's selection probability in the next stage.

In Stage 2, the importance of all candidate predictors undergoes further evaluation through a second round of the rPQL applications to a new set of bootstrapped samples. The predictors with high values in the index, obtained from the first stage, are more likely rolled into the new subsets of predictors within each bootstrapped sample. As noted previously, the rPQL could yield biased estimates, particularly when the number of observations within each cluster is small relative to the number of clusters (X. Lin & Breslow, 1996). To mitigate such potential biases, we employ an unbiased estimation with restricted maximum likelihood (Hui et al., 2017) immediately after implementing rPQL in each bootstrapped sample. That is to specify a GLMM with the fixed and random effects selected by rPQL, and use the maximum likelihood estimation to obtain the estimates (i.e., β and D) for these selected variables. All estimates are then concatenated across all bootstrapped samples to form two matrices. One is a $B' \times p$ estimate for the fixed effects, and the other is a $B' \times q$ estimate for the random effects (where p and q, respectively, represent the number of candidate predictors for the fixed and random effects, and B' is the number of bootstrapped samples in the second stage). Lastly, a selection criterion, stated later, is used to select the final set of important predictors.

A few technical specifics should be noted for random rPQL. First, to ensure selection stability, the number of bootstrapped samples generated in each stage is determined by using the method proposed by Kim et al. (2019).[3] Second, at the end of Stage 1, the index of predictors is computed by calculating the absolute values of estimates and then calculating the arithmetic mean across those absolute values throughout all bootstrapped samples, which is different from an existing approach that reverses the order of this computation (Wang et al., 2011). This adjustment prevents potential shrinkage of the index at the end of Stage 1, helping to avoid the underestimation of selection probability for predictors in Stage 2, especially when estimates exhibit opposing signs in different bootstrapping due to multicollinearity among predictors. The detailed description of the random rPQL procedure is provided in the online supplemental materials and also shown in Figure 1.

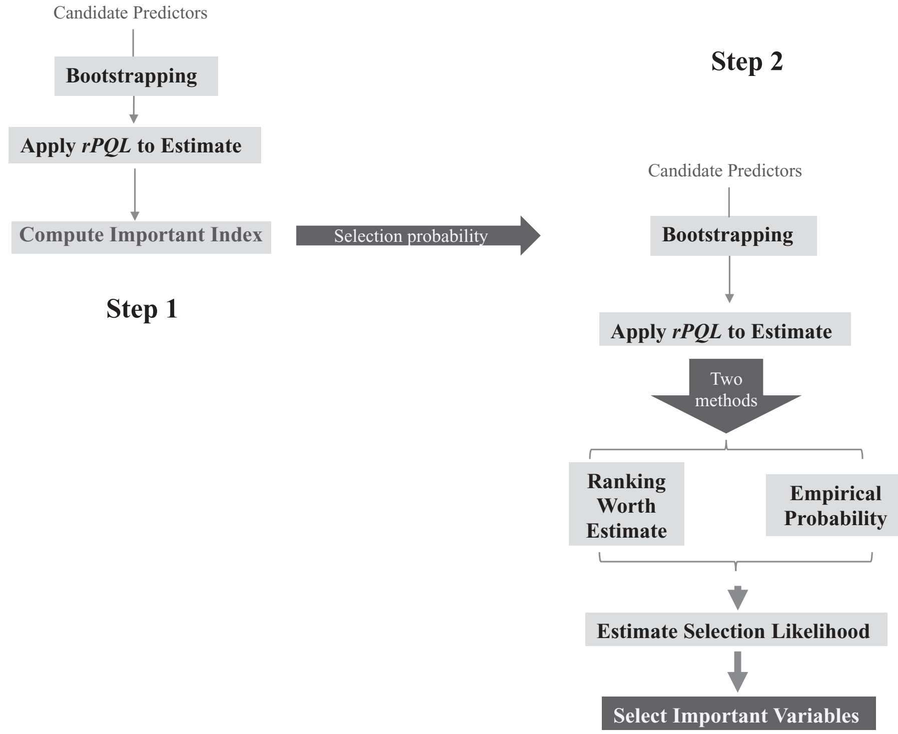
Criterion for Selection of Important Variables

A selection criterion is a process to estimate the selection probability. It is employed to identify the ultimate set of important predictors across all bootstrap samples at the end of Stage 2. One selection criterion known as empirical probability has been used under the context of GLMs (Liu et al., 2016). It defines the variables as important if they have high frequencies of nonzero estimates across all bootstrapped samples. Specifically, the empirical probability criterion first transforms the estimate matrices obtained in Stage 2 into those with dichotomic values; that is, replacing any nonzero estimates with a value of one and zero estimates with a value of zero. The occurrences of a predictor yielding nonzero estimates across all bootstrapped samples divided by the total number of bootstrapped samples are defined as that predictor's empirical probability (Liu et al., 2016; Meinshausen & Bühlmann, 2010; Wang et al., 2011). Predictors with high empirical probability are regarded as important.

The empirical probability criterion assumes that regularization can accurately shrink the coefficients of nonimportant variables to zero within each bootstrapped sample. However, this assumption could be violated if the regularization approach produces false-positive selections. It typically occurs when the penalty parameter (λ) is set to be too weak or the sample size is too small. The choice of λ determines the accuracy with which the regularization method can distinguish important variables from noise. With a weak λ, the coefficients of nonimportant variables may not shrink to zero. To prevent choosing a weak λ, the random rPQL adopts the information criteria proposed by Hui et al. (2017) to optimize the penalty parameter (λ) within a predefined set of potential λ values. This method has been shown to be consistent and efficient within a single data set. However, its adaptability across numerous bootstrapped samples remains uncertain. Due to random draws, a set of potential λ values used to optimize λ within one bootstrapped sample may not include the equally optimal one for another sample, which might increase the likelihood of selection errors. Furthermore, as the sample size decreases, regularization approaches are more likely to produce false-positive selections, even with an optimal λ (Kirkland et al., 2015). This results in the coefficients of nonimportant predictors not shrinking to zero, which likely inflates their importance under the empirical probability criterion.

[3] The number of bootstrapped samples were computed by $B \geq \frac{z^2_{\alpha/2}L(1-L)}{d^2}$, where $z_{\alpha/2}$ is 1.96 at $\alpha = .05$, d is a predefined estimate error, and L is a proportion of predictors in the subset over the total number of predictors in each bootstrap.

Figure 1
Procedures for Random rPQL



Note. rPQL = regularized penalized quasi-likelihood.

To improve the selection accuracy, we introduce a new criterion named ranking worth estimation into the variable selection in the context of GLMMs. This criterion, derived from the extended Plackett-Luce model (Firth et al., 2019; Turner et al., 2020), is used to assess the importance of variables. Under the proposed variable selection algorithm, this criterion determines variables' importance based on their likelihood of ranking at the top across all bootstrapped samples. Specifically, ranking worth estimation ranks each candidate predictor in reference to the magnitude of its estimates at every individual row of the aforementioned estimate matrices ($B' \times p$ and $B' \times q$, respectively). The predictors associated with larger nonzero estimates are placed with higher ranks, while those with zero estimates are ranked at the lowest. The predictors that failed to be included in the subsets at each bootstrapping are treated as missing values in the rankings. Notably, the ranking worth estimation is applied to the two matrices produced in Stage 2. Two ranking matrices are thus generated by separately ranking the importance of all predictors based on the fixed and random effects. Then, the extended Plackett-Luce model used to produce the ranking worth estimation is applied separately to the two ranking matrices to obtain the probability of each predictor being ranked at first within their corresponding lists of candidate predictors. Predictors with higher probabilities are regarded to be more important.

The Plackett-Luce model has been widely applied in other fields such as political science (Gormley & Murphy, 2008), agriculture (Simko & Pechenick, 2010), economics (Njenga et al., 2018), and recommender systems (Diaz et al., 2020). It serves as a useful tool to model the probability of a specific rank ordering for items (Finch, 2022). The item's worth is the focal model parameter, which represents the importance of the items in the rankings and can be estimated through maximum likelihood. We integrate the Plackett-Luce model into our novel variable selection algorithm, using its key parameter as a new selection criterion. This approach is expected to be superior to empirical probability because it weighs those truly important predictors against noise peers by accounting for the magnitude of the estimates. When the sample size is small and/or λ is weak, the magnitude of the estimates for nonimportant predictors could be small but not exactly zero. In this case, the empirical probability will mistakenly regard the nonimportant predictors as important. In contrast, the ranking worth estimation assigns lower ranks to such nonimportant predictors, thereby decreasing the probability of nonimportant variables being ranked at the top.

The proposed random rPQL alongside ranking worth estimation could contribute to variable selection in GLMMs in a few ways. First, through utilizing resampling techniques, this method could effectively address the challenges of multicollinearity by breaking up the correlations among variables. Second, the number of predictors involved in a bootstrapped sample can be manipulated to be much smaller than the number of available observations within a cluster. This unique feature would benefit the feasibility and efficiency in variable selection when $p > n$. Third, the repeated estimations of GLMMs in the bootstrapped samples at both stages would improve the consistency and stability of selection results. Last, the selection criterion of ranking worth estimation is expected to increase the selection accuracy, as it accounts for the magnitude of the estimates rather than solely the status of zero or nonzero.

In the following sections, we conducted two simulation studies to examine the performance of the proposed random rPQL technique in conjunction with the ranking worth estimation for variable selection in GLMMs. As a comparison, empirical probability was also used as an alternative selection criterion in both studies. Both studies used the continuous outcomes but differed in the relationship between the number of observations within each cluster (n_j) and the number of candidate variables (p), with $n_j > p$ in Study 1 and $n_j < p$ in Study 2. We also ran an additional simulation study with binary outcomes. Appendix provides the details and results of this study.

Simulation Study 1: The Performance of Random rPQL When $n_j > p$

Data Generation

Study 1 was characteristic of an ideal situation where the number of observations (n_j) exceeded the number of candidate variables (p) within each cluster. The population model, a GLMM expressed by Equation 2 with $X_j = Z_j$ and $\varepsilon_j \sim N(0, 1)$, was used to generate continuous data with 500 replications per condition. The first column of X_j and Z_j was constructed with ones as the basis coefficients for intercepts. The remaining elements of X_j and Z_j were generated from a multivariate normal (MVN) distribution $\sim$MVN$(0, \Sigma)$, where Σ was chosen based on the preset multicollinearity status.

Variables in X_j and Z_j were the candidates for the fixed effects and random effects, respectively. The total number of variables in X_j and Z_j was set to be equal, that is, $p = q = 10$ or $p = q = 30$. The five coefficients (β_1 to β_5) and three variances in covariance matrix D were designed not equal to zero. Thus, the five fixed effects (X_1 to X_5) and three random effects (Z_1 to Z_3) were expected to be important. The rest of X_j and Z_j were designed to be nonimportant variables, so their corresponding coefficients (both the fixed and random effects) and variance components were set to zero. The proportions of the important variables for the fixed effects were set at 5/10 and 5/30, corresponding to $p = q = 10$ or $p = q = 30$, respectively. The proportions were set at 3/10 and 3/30 for the random effects.

The parameter values of the GLMM were chosen based on prior simulation studies on variable selection (Bondell et al., 2010; Y. Fan & Li, 2012; Hui et al., 2015; B. Lin et al., 2013; Lu et al., 2024). The manipulated factors included group sizes, within-group size, number of variables, effect sizes of β, and multicollinearity. Specifically, we generated data using three effect sizes of β, that is, small, large, and mixed:

$$\begin{aligned} \beta_{\text{small}} &= (0.1, 0.5, 0.4, 0.3, 0.2, 0, \ldots, 0), \\ \beta_{\text{large}} &= (0.1, 0.8, 0.7, 0.6, 0.5, 0, \ldots, 0), \\ \beta_{\text{mixed}} &= (0.1, 0.2, -0.2, -0.9, 0.9, 0, \ldots, 0). \end{aligned} \tag{3}$$

Noted that the first element of β was the intercept with the same value of 0.1. We adopted the covariance matrix used in prior research for the true random effects (Bondell et al., 2010; Y. Fan & Li, 2012; Hui et al., 2015; B. Lin et al., 2013):

$$D = \begin{bmatrix} 9 & 4.8 & 0.6 & \ldots & 0 \\ 4.8 & 4 & 1 & \ldots & 0 \\ 0.6 & 1 & 1 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & 0 \\ 0 & 0 & 0 & \ldots & 0 \end{bmatrix}. \tag{4}$$

This was set to be the same across all conditions. The within-group size was designed to be $n = 30$ or 50, representing medium to large samples in social sciences (Maas & Hox, 2005; Meinck & Vandenplas, 2012). The group sizes were set at $m = 15$, 30, 50, and 100, reflecting a small to large number of groups (Kreft & de Leeuw, 1998; Krull & MacKinnon, 2001; Maas & Hox, 2005). Under the multicollinearity conditions, we applied a high correlation of $\rho = .9$ to either the important variables (x_2, x_3), (x_4, x_5), or the nonimportant variables (x_7, x_8; Kim et al., 2019). The correlations among all other variables were equivalently set to $\rho = .05$.

Variable Selection With Random rPQL

The random rPQL was used to select variables when building a GLMM with simulated data. A predefined range of penalty parameters λ was $[.1, 1 \times 10^{-10}]$ for rPQL on Stage 1, and $[1, 1 \times 10^{-4}]$ on Stage 2. The optimal penalty parameter λ for each bootstrapping was selected by the information criterion developed by Hui et al. (2015; see footnote 2). In each bootstrapped sample, the debiased estimation was used to estimate the fixed and random effects with nonzero coefficients after those effects were selected by rPQL (Hui et al., 2015), which ensured the index of the variables was calculated based on the debiased estimates. The two selection criteria, that is, ranking worth estimation and empirical probability, were used to identify the final set of important predictors under all conditions. A selection was treated to be correct when all important fixed and random effects produced higher empirical probability or ranking worth estimation compared to their nonimportant counterparts. Otherwise, the selection was regarded as a false positive.

Results

Power rate is defined as the percentage of the replicates that correctly selected all important predictors and correctly excluded all nonimportant predictors under a specific condition. A higher power rate indicates a greater accuracy of variable selection. In our simulation studies, we calculated power rates for the fixed effects, random effects, and the entire model.

Overall, random rPQL exhibited excellent performance with consistently high average power rates of 0.98 ($SD = 0.04$) for the fixed effects, 0.99 ($SD = 0.01$) for the random effects, and 0.98 ($SD =$ 0.04) for the overall model. The two criteria were equally competitive as both produced the same average power rates of 0.98 for the fixed effects, 0.99 for the random effects, and 0.98 for the overall model.

Table 1 summarizes the average power rates under the two selection criteria at each level of the simulation factors. For the fixed effects and the overall model, the higher power rates were associated with larger group sizes, greater within-group sizes, larger effect sizes, and smaller number of variables. For the random effects, on the other hand, the power rates were consistently high ($\geq$0.99) across all manipulated factors. This indicated that the changes in power rate for the overall model were mainly associated with the fixed effects.

A further inspection revealed that random rPQL was able to successfully select variables even with a small to moderate size of groups; for instance, the average power rates were sufficiently high (0.96) with only 15 groups. In addition, when the number of predictors increased from 10 to 30, the average power rates remained high

Table 1
Average Power Rate for Fixed Effect, Random Effect, and Combining Both Effects for Each Level of Design Factors

	Fixed effect		Random effect		Combining both effects	
Factor	RWE	EP	RWE	EP	RWE	EP
Group size						
15	0.96	.96	0.99	.99	0.96	.96
30	0.98	.98	0.99	.99	0.98	.98
50	0.99	.99	0.99	.99	0.99	.99
100	0.99	.99	0.99	.99	0.99	.99
Within-group size						
30	0.98	.97	0.99	.99	0.97	.97
50	0.99	.99	0.99	.99	0.99	.98
Number of predictors						
10	0.99	.99	0.99	1.00	0.99	.99
30	0.97	.97	0.99	.99	0.97	.97
Fixed-effect effect size						
Small	0.97	.97	0.99	.99	0.97	.97
Mixed	0.98	.98	0.99	.99	0.98	.97
Large	0.99	.99	0.99	.99	0.99	.99
Multicollinearity						
No	0.98	.97	0.99	.99	0.97	.97
Yes	0.99	.99	0.99	.99	0.99	.99
Overall	0.98	.98	0.99	.99	0.98	.98

Note. RWE = ranking worth estimation; EP = empirical probability.

from 0.99 ($SD = 0.02$) to 0.97 ($SD = 0.04$). A negligible drop was observed in the power rates from 0.98 ($SD = 0.03$) to 0.94 ($SD = 0.08$), even when the group size was as small as 15. Overall, random rPQL demonstrated a high accuracy of variable selection, even under unideal conditions where a large number of predictors was combined with a low to medium group size.

Furthermore, it is expected that decreasing the effect sizes of the fixed effects would negatively impact the selection accuracy (B. Lin et al., 2013); however, this was not evident in our study. For instance, the average power rate dropped slightly from 0.99 ($SD = 0.01$) to 0.96 ($SD = 0.04$) when the effect sizes were changed from large to small. Moreover, under the conditions with small effect sizes, the power rates decreased from 0.99 ($SD = 0.01$) to 0.94 ($SD = 0.06$) when the number of groups decreased greatly from 100 to 15. Thus, our proposed method withstood a high accuracy of variable selection despite the effect size of the fixed effects being small. Finally, the multicollinearity was not detrimental to the selection accuracy. Instead, the power rate in the conditions with multicollinearity ($M = 0.99$, $SD = 0.01$) was slightly better than that without multicollinearity ($M = 0.97$, $SD = 0.05$).

To summarize, when candidate predictors did not outnumber observations within clusters, the random rPQL method was able to select important variables with remarkably high accuracy and stability across a range of data conditions. The average power rates were consistently high (>0.90) with an ignorable variability across conditions (<0.10).

Simulation Study 2: The Performance of Random rPQL When $n_j < p$

Data Generation and Variable Selection

The second study was characterized by an equal or greater number of candidate predictors (p) than observations (n_j) within each cluster. Such a scenario is common in longitudinal research where data are collected from a limited number of waves while the candidate predictors outnumber the waves. Study 2 intended to explore the extent to which random rPQL could accurately select important predictors.

Data were generated by the same model (Equation 1) with almost identical conditions to those in Study 1, except for the factors of within-group size and group size. The within-group size was designed to be $n = 5$ or 10, resembling the limited number of waves employed in typical longitudinal research. The group sizes were set at $m = 50$, 100. The two selection criteria, empirical probability and ranking worth estimation, were also used. Power rates were used to evaluate the performance of random rPQL on selecting the fixed effects, the random effects, and the whole model.

Results

Overall, random rPQL produced a high level of accuracy in variable selections. The average power rates were 0.87 ($SD = 0.20$) for the fixed effects, 0.97 ($SD = 0.07$) for the random effects, and 0.85 ($SD = 0.22$) for the overall model. Regardless of the selection criteria, higher power rates were associated with larger within-group sizes, greater group sizes, and larger fixed effect sizes but with a smaller number of variables. In addition, the ranking worth estimation outperformed the empirical probability for both fixed and random effects. It yielded the average power rate of 0.89 ($SD = 0.18$) for the overall model, higher than the empirical probability's power rate of 0.82 ($SD = 0.24$; Table 2). Considering the main purpose of random rPQL was to select the correct models overall, we used the average power rates on the overall model to illustrate the impact of each factor.

First, the increment of within-group size from $n = 5$ to $n = 10$ substantially improved the selection accuracy when the within-group size was set to be smaller than the number of variables. The power rate increased from 0.76 ($SD = 0.26$) to 0.95 ($SD = 0.08$). Furthermore, the within-group size was associated with the selection criterion applied. With $n = 10$, the power rates remained high regardless of selection criteria (i.e., 0.96 by empirical probability and 0.95 by ranking worth estimation). However, with $n = 5$, the power rate of ranking worth estimation was 0.83 ($SD = 0.23$), remarkably higher than 0.69 ($SD = 0.28$) of empirical probability.

Second, the large group size could improve selection accuracy when dealing with a limited within-group size. When the group size grew from 50 to 100, the power rate increased by about 10.7% conditionally on only five observations per group (i.e., $n = 5$). It was much larger than 4%, which was conditional on the within-group size of 10. Additionally, we found that a large group size (from $m = 50$ to $m = 100$) may also benefit the selection when the number of variables was large, such as $p = 30$. The power rate increased by up to 12%.

Like Study 1, decreasing the effect size of the fixed effects could negatively impact the selection accuracy from 0.95 ($SD = 0.1$) to 0.83 ($SD = 0.2$). Unlike Study 1, ranking worth estimation became superior when the coefficients were small. It yielded a power rate of 0.89 ($SD = 0.16$), much higher than the empirical probability of 0.78 ($SD = 0.23$).

The stability of random rPQL in selection remained remarkable, even with an increased number of variables. As the number of variables (p or q) grew, the proportion of important variables among all candidate variables dropped (e.g., from 83% to 50%). It made

Table 2
Average Power Rate for Fixed Effect, Random Effect, and Combining Both Effects for Each Level of Design Factors

Factor	Fixed effect RWE	Fixed effect EP	Random effect RWE	Random effect EP	Combining both effects RWE	Combining both effects EP
Group size						
50	0.87	.81	0.97	.93	0.86	.78
100	0.94	.87	0.98	.98	0.93	.86
Within-group size						
5	0.85	.73	0.97	.92	0.83	.69
10	0.97	.95	0.99	.99	0.96	.95
Number of predictors						
10	0.93	.87	0.99	.97	0.92	.85
30	0.88	.82	0.97	.93	0.87	.79
Fixed-effect effect size						
Small	0.91	.80	0.97	.95	0.89	.78
Mixed	0.82	.75	0.99	.96	0.82	.74
Large	0.99	.97	0.97	.95	0.97	.94
Multicollinearity						
No	0.83	.75	0.96	.92	0.80	.71
Yes	0.99	.94	0.99	.99	0.98	.93
Total	0.91	.84	0.98	.95	0.89	.82

Note. RWE = ranking worth estimation; EP = empirical probability.

distinguishing the important variables from the rest of the noises more challenging, especially with limited within-group size. However, random rPQL consistently provided a high accuracy of 0.89 ($SD = 0.18$) at $p = 10$ and 0.83 ($SD = 0.25$) at $p = 30$. This exceptional accuracy persisted regardless of the within-group size. Even with only five observations within a group, the difference in power rates was nearly negligible between $p = 10$ at 0.79 ($SD = 0.22$) and $p = 30$ at 0.72 ($SD = 0.30$).

Over and beyond the qualities obtained above, random rPQL performed surprisingly well in selecting all highly correlated and important predictors. The power rate of 0.96 ($SD = 0.26$) was excellent and notably much better than 0.75 ($SD = 0.26$) from the conditions without multicollinearity. Moreover, we found two selection criteria that performed distinguishably between multicollinearity on and off. The empirical probability greatly increased the power rate from 0.71 ($SD = 0.29$) to 0.93 ($SD = 0.09$), as multicollinearity was introduced in the data. In comparison, ranking worth estimation did not improve that much, from 0.8 ($SD = 0.22$) without to 0.98 ($SD = 0.02$) with multicollinearity.

If we further examined the interactions between multicollinearity and other factors, power rates were found to be improved under conditions with small group size, small within-group size, small effect sizes of fixed effect coefficients, and a large number of variables. For instance, multicollinearity increased 33.7% of accuracy in conditions with only five observations in a group ($n = 5$).

Finally, we found that the computation time of random rPQL was affected by sample size (including group size and within-group size) and the number of bootstrapped samples used. Running on a computer equipped with an i7 processor, clock speed of 2.9 GHz, and 32 GB RAM, it took about 20 min for a condition where the group size was 300, within-group size was 5, and 400 times bootstrapping. In reference to the longitudinal research, this setting is typical and feasible. It should be noted that increasing either the group size or within-group size would exponentially increase the computation time.

In summary, Study 2 clarified the high qualifications that random rPQL earned on variable selection accuracy. It showed a powerful capability to identify important variables from the data in which the number of candidate predictors exceeds the sample size at the cluster level. The new proposed ranking worth estimation provided better alternative criterion under various conditions.

Empirical Study

We demonstrated an application of random rPQL to select important fixed and random effects in a GLMM by using the data collected from the U.K. Millennium Cohort Study. Millennium Cohort Study is a large, ongoing longitudinal study tracking a cohort including about 19,500 children (Joshi & Fitzsimons, 2016). To date, the children and their families have been interviewed seven times consecutively at the ages of 9 months and 3, 5, 7, 11, 14, and 17 years.

In this empirical study, the outcome variable was derived from the Strengths and Difficulties Questionnaire (SDQ; R. Goodman, 1997). The SDQ contains 25 items measuring five children's behavior attributes, including emotional symptoms, conduct symptoms, hyperactivity/inattention, peer relationship problems, and prosocial behaviors. Since prosocial behaviors are conceptually distinct from other psychological difficulties (A. Goodman et al., 2010), items of prosocial behaviors were excluded from the computation of the outcome variable. All items of SDQ were responded using a Likert scale from *not true* (Scored 0), *somewhat true* (Score 1), to *certainly true* (Score 2). Higher scores indicate more severe levels of psychological and behavioral problems. We chose 12 variables as the candidate predictors (Table 3), most of which were found to be associated with the outcome SDQ in previous studies (Fitzsimons & Villadsen, 2019; Harkness et al., 2020; Moulton, et al., 2021; Noonan et al., 2018; Wickham, et al., 2017).

Our data consisted of self-reports from a sample of 1,126 mothers who had attended at least five assessments. For each mother, we used the data collected when their children were at ages of 3, 5, 7, 11, and 14, as parents' SDQ reports were available at these time points. The intraclass correlation coefficients indicated that 11% of the total variability in psychological difficulties was attributable to between-person differences, while 89% was due to within-person changes over time. Additionally, among all the candidate predictors, the variables "Income-Weighted Quantiles" I (X_4) and II (X_{12}) were highly correlated ($r_{X_4X_{12}} = .88$, $p < .001$), serving as an example of potential multicollinearity in variable selection. Considering the computational demands from running variable selection on a large sample, we randomly selected 500 out of 1,126 mothers to illustrate the application of random rPQL. Approximately 41.4% of the participants had missing data. We used a nonparametric approach named biobjective k-nearest-neighbor-based imputation to deal with the missingness. Previous research has shown that this method performs competitively with multiple imputation for clustered data (Cubillos et al., 2022).

To illustrate the entire process of applying variable selection procedure to this data set, we followed a framework commonly used in regularization research (e.g., Wang, et al., 2011; Zou & Hastie, 2005). The data were first randomly split into a training set (70% of the data, sample size = 350) and a testing set (30% of the data, sample size = 150). We then respectively applied random rPQL to the training set for variable selection, and estimated the

Table 3
Descriptions of Candidates Predictors

Predicator label	Predictor name	Level of predictor
X_0	Intercept	
X_1	Wave	Wave 1, Wave 2, Wave 3, Wave 4, Wave 5
X_2	KESS score (psychological distress)	Range = 0–18, higher score means higher distress
X_3	Parent's highest academic level	Level 1–Level 5; higher level means higher education
X_4	Income-Weighted Quantile I (based on a single country)	Level 1–Level 5; 1 = *lowest quantiles*, 5 = *highest quantiles*
X_5	Natural father at home	0 = *natural father not in household*, 1 = *natural father in household*
X_6	Parents/careers in household	1 = *two parents/careers in household*, 2 = *one parent/care in household*
X_7	Number of people in household	
X_8	Relationship between parents/careers	1 = *married*, 2 = *cohabiting*, 3 = *neither*
X_9	Mother in work	0 = *not in work*, 1 = *in work*
X_{10}	Father in work	0 = *not in work*, 1 = *in work*
X_{11}	Household status	1 = *live in a house that is owned/mortgaged*, 2 = *rent*, 3 = *live for free*
X_{12}	Income-Weighted Quantile II (based on United Kingdom)	Level 1–Level 5; 1 = *lowest quantiles*, 5 = *highest quantiles*

Note. KESS = Kessler Psychological Distress Scale.

parameters of the selected predictors by fitting a GLMM to the testing set.

Specifically, we used a linear mixed model to predict the long-term changes of psychological difficulties between 3 and 14 years of age. To identify important predictors out of all 12 candidates, we applied random rPQL algorithm with the same predefined ranges of penalty parameters λ from the simulation studies and used the ranking worth estimation criterion for variable selection. All 12 variables were considered as candidate predictors for both fixed and random effects. Limited by only five waves of measurements for each variable, a maximum of four predictors could be randomly subset from the pool of 12 candidates for each bootstrapping. Consequently, in each bootstrapped sample on both Stage 1 and Stage 2, four predictors (one out of three of all candidates) were randomly drawn and included in a linear mixed model as the fixed effects. Similarly, another four (which may or may not be the same as the four fixed effects) predictors were also randomly drawn and included as the random effects. The bootstrapping set to be 400 for both stages, calculated by the equation proposed by Kim et al. (2019). Given the selection probability produced by the ranking worth estimation could be very small for those nonimportant variables, Figure 2 provides the log-transformed values for a better visualization.

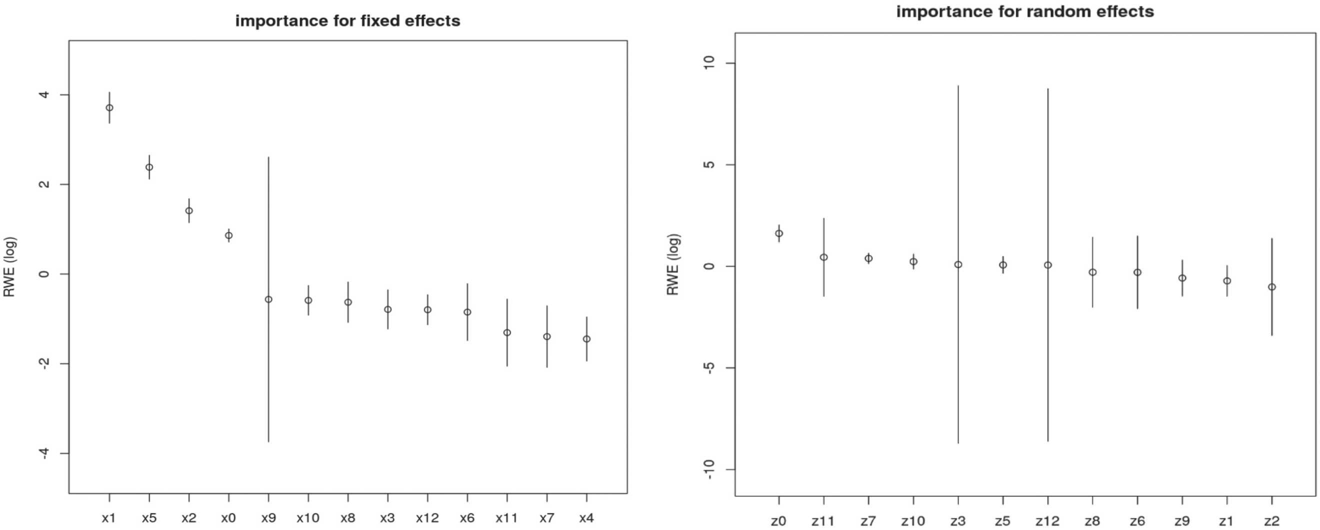
We evaluated the importance of each predictor through examining the fixed and random effects separately. In terms of the fixed effects, the Kessler Psychological Distress Scale score, natural father at home, and wave, these three variables (X_2, X_5, X_1) and the intercept (X_0) should be selected due to their high values in selection probability, while the other nine variables (X_3, X_4, X_6, X_7, X_8, X_9, X_{10}, X_{11}, X_{12}) should be excluded. The ranking worth estimation on the random effects showed that only Z_0 (intercept) yielded a relatively higher value in selection probability, while other predictors did not. Thus, Z_0 (intercept) was selected. Finally, we fit an analytic model with the selected variables on the testing set of data. Table 4 provides the results, indicating that children from disadvantaged families suffered a more severe level of psychological behavior problems as they grew from a young age to a teenager. Moreover, the father figure's absence from the household was harmful for child mental health, and individual's high distress levels could be another important factor related to the high SDQ.

Discussion

In the era of big data, the wealth of available information enables researchers in social and behavioral sciences to collect a wide range of variables. This abundance supports scientific advancements, particularly when theoretical frameworks are evolving—with some theoretical justification but not yet fully confirmatory. In such cases, hypotheses tend to be broader and more complex, often involving numerous variables in data sets. For example, a study may include theoretically justified variables without prior examination of their interrelationships or may integrate established predictors with additional exploratory variables to better capture the complexities of human behaviors (Jacobucci et al., 2023). In this context, variable selection is valuable in streamlining the candidate predictors, simplifying models, enhancing computational efficiency, and ensuring that research questions are effectively addressed (Kerssens-van Drongelen, 2001; Sanbonmatsu et al., 2025).

From a statistical modeling perspective, variable selection serves as a critical preprocessing step. It focuses on those most relevant while discarding less important variables, so that it mitigates the risk of constructing overly complex models and prevents model overfitting. Specifically, in the context of clustered data analyzed via GLMMs, variable selection supports accurately and simultaneously identifying important fixed and random effects. Furthermore, it safeguards against the oversimplification of random effects, particularly when between-group variability is nonnegligible. As important random effects fail to be included in building GLMMs, underestimating fixed effects' standard errors is more likely to occur, increasing the Type I errors in the hypothesis test. It also aims to prevent overly complex random effects involved

Figure 2
Log-Transferred Estimates for the Ranking Worth for Candidate Fixed and Random Effects



(a) Ranking worth for candidate fixed effect

(b) Ranking worth for candidate random effect

Note. RWE = ranking worth estimation.

in model building, where GLMMs may struggle to converge (Matuschek et al., 2017; Vergouwe et al., 2010). However, achieving this goal in the context of GLMM is difficult due to three challenges—the high computational costs, the outnumber of variables within a cluster, and multicollinearity.

In response, the current study proposed a novel method, random rPQL. This approach integrates rPQL into resampling processes, offering a solution for simultaneously selecting fixed and random effects with higher computational efficiency and accuracy. By incorporating resampling techniques, random rPQL extends rPQL's applicability to handle high-dimensional predictor sets that exceed the available data size within a cluster. Furthermore, random rPQL demonstrates the capability to identify all important, correlated predictors, thereby enhancing the stability of variable selection. It also addresses a limitation observed in LASSO/adapted LASSO methods (used in rPQL), which may lead to the selection of nonidentical subsets of highly correlated important variables across different data sets.

Our two simulation studies showed the extraordinary performance of random rPQL in two scenarios: the number of predictors is relatively larger compared to the group sizes (Study 1), and the number of observations within each cluster exceeds the predictors (Study 2). Furthermore, the empirical study demonstrated that the variable selected by the random rPQL is meaningful and interpretable in practice. Notably, we refrained from comparing random rPQL with other GLMMs variable selection methods, as they are either unable to handle scenarios where the number of candidate predictors exceeds the observations within each cluster or are hindered by computational intensity (e.g., Y. Fan & Li, 2012).

Our secondary contribution is introducing a novel criterion, the ranking worth estimation. Across diverse simulation conditions, this criterion has proven to be superior to the previously used empirical probability criterion. The main difference between the two lies in their interpretation of variable importance. The empirical probability criterion converts estimates to binary, categorizing a variable's importance to be dichotomy. In contrast, the ranking worth estimation treats estimates as ranking data, reflecting the relative nature of variable selection, where importance depends on the extent of correlation between the predictor and the outcome of interest. By treating estimates as ranking data, the ranking worth estimation allows the Plackett-Luce model to provide the probability of each variable ranking at the top.

Furthermore, the ranking worth estimation is more robust in the process of regularization integrated with the resampling approach. Typically, in a single data set, selection accuracy depends on the choice of the penalty parameter in regularization (e.g., Hui et al., 2017; Sun et al., 2013). Methods such as cross-validation and information criteria (e.g., Bayesian information criterion [BIC] or Akaike information criterion [AIC]) are commonly used in GLM to determine the

Table 4
General Linear Mixed Model Results for Predicting SDQ

Predictor	Estimate	*SE*	*df*	*t*	*p*
Fixed effect					
Intercept	0.41***	0.03	553	12.96	<.001
KESS score (distress)	0.08**	0.02	745	4.35	.002
Natural father in household (0 = *not in household*)	−0.09***	0.03	262	−3.51	<.001
Wave	0.04***	0.01	681	4.21	<.001
Variance components					
Var (Int)	0.01	0.003		2.54[a]	.006
Var (Residual)	0.08	0.004		17.18[a]	<.001

Note. SDQ = Strengths and Difficulties Questionnaire; KESS = Kessler Psychological Distress Scale; Var (Int) = variance (Intercept); Var (Residual) = variance (Residual).
[a] Test for variance components is z test.
** $p < .01$. *** $p < .001$.

optimal penalty. Uniquely, the current study involves multiple bootstrapped samples, where selection accuracy depends not only on the penalty choice but also on the selection criterion. When a penalty can be optimized, selections based on either empirical probability or ranking worth estimation should not differ significantly. However, when a penalty cannot be easily optimized, the advantage of ranking worth estimation becomes more apparent. For example, in the context of GLMMs, it is complicated to integrate penalty optimization into resampling techniques. A predefined range of penalty values to optimize λ for rPQL at one sample may not be sufficient for every bootstrapped sample. As both observations and predictor subsets vary across bootstrapped samples, the optimal penalty range in one sample may not be optimal in another, possibly increasing false-positive or false-negative selections. Especially with a limited sample size, optimizing the penalty is more challenging, leading to regularization to produce high false-positive selections (Leng et al., 2006; Zou et al., 2007). That is, the estimates of nonimportant variables may fail to shrink to zero (but a very small value instead), and the empirical probability criterion, in this case, would falsely include them. By contrast, the ranking worth estimation is more robust by taking the small estimates into account and decreasing the likelihood of false-positive or false-negative selections across all bootstrapped samples.

Limitations and Future Directions

In summary, the proposed random rPQL, a variable selection method, shows the capability of effectively identifying important fixed and random effects in GLMMs. However, the limitations are inevitable but can be the direction for future research. First, we did not provide a predetermined (range of) threshold (π_{thr}) on the selection probability produced by ranking worth estimation or empirical probability. According to previous studies (Wang et al., 2011), variables' selection probability above and beyond a certain threshold were expected to be selected. Several practices suggested to calculate such threshold by $\pi_{\text{thr}} = 1/n$ (Wang et al., 2011; n is the sample size of training data), or using cross-validation (Liu et al., 2016). Unfortunately, no well-established theories have been proposed to support any of them. In addition, those practices were limited only in GLM. For GLMMs, however, preestablishing a threshold seems to be less applicable, due to the infeasibility in cross-validation to calculate random coefficients (labeled as u_j in Equation 2). Furthermore, the sum of the worth values generated by ranking worth estimation should equal one among all candidate predictors. It quantifies the variable's level of importance in relation to its competitors rather than a predefined confirmatory value. Thus, seeking such a threshold is neither meaningful nor practical in the application of ranking worth estimation. Nonetheless, researchers may require empirical guidelines for variable selection based on their ranking orders. In addition to the visual approach (shown in the empirical study), we provided, future studies should explore statistical solutions. The second limitation relates to the ranking worth estimation, which is solely calculated using the Plackett-Luce model. It is worth noting that alternative methods in ranking and choice modeling exist, such as the Bradley–Terry model (Bradley & Terry, 1952), the distance-based ranking models (Lee & Yu, 2010), et al. In future studies, it would be beneficial to investigate the performance of these alternative models. Third, this research assumes normality on the random effects, and only employed continuous (Studies 1 and 2) and binary (Appendix) outcome variables. Future research should explore the performance of random rPQL when normality assumptions are violated, or when other types of outcome variables are presented.

Furthermore, to deal with missing data (i.e., in our empirical study), future research should explore the appropriate combination of regularization techniques and more advanced methods, such as multiple imputation (Gunn et al., 2023). Finally, it is important to note that random rPQL serves as a tool for variable selection and not for the simultaneous estimation of parameters. Even when important variables are identified, it remains necessary to employ unbiased estimators like restricted maximum likelihood for hypothesis testing in the analytic stage. In such instances, where multiple correlated variables are involved, researchers need to follow the methodological guidance on handling multicollinearity outlined in the relevant literature.

Suggestion for Applications

The proposed random rPQL represents a novel application of regularization techniques for variable selection within the framework of GLMMs. It has been specifically tailored to address scenarios where the sample size is limited, but the number of predictors of interest is large. Our studies have demonstrated its exceptional performance, even when dealing with restricted sample sizes. However, it is essential to recognize that the primary focus of this method is on selecting important variables rather than estimating model parameters.

Furthermore, the application of variable selection should not deviate from theoretical guidelines. A primary goal in social and behavioral sciences is to build models to test, interpret, and confirm theoretical frameworks. Relying solely on data-driven techniques, such as variable selection, without any theoretical grounding can lead to models being less interpretable or even conceptually meaningless.

Finally, in order to have a better application of random rPQL in social and behavioral research, we offered several recommendations. First, prioritize ranking worth estimation criterion over empirical probability. Use the visualized estimates of worth parameters (such as Figure 2) to make decisions on important variables, which rely on the patterns where the average estimates consistently exhibit higher values with minimum dispersion. It indicates that their rankings are more frequent at the top compared to others, whose estimates either continually remain low or frequently fluctuate, resulting in great variabilities. Variables that fall in between may indicate a moderate association with the outcome but do not carry as much importance as their peers. For these intermediate cases, users need to take both theories and variable selection results into account and carefully make decisions. Second, a large within-group size helps to improve variable selection in GLMMs. However, it might be challenging in practice (e.g., longitudinal studies). Alternatively, increasing group size can improve selection accuracy if obtaining a sufficiently large within-group size is not feasible. Third, random rPQL can identify both highly correlated and important variables, primarily aimed at avoiding the inadvertent masking of multicollinearity. While multicollinearity may benefit in accurately selecting important variables, it is still a challenge in hypothesis testing. The subsequent decisions regarding which collinear variables to include in the final models are not within the random rPQL primary scope. Once the selection is complete, users should adhere to the guidance of statistical protocols to handle multicollinearity.

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Appendix

Simulation Study 3: The Performance of Random Regularized Penalized Quasi-Likelihood (rPQL) When the Outcome Is Binary

Data Generation and Variable Selection

In Study 3, the outcome variable was generated to be binary. This study employed the same set of candidate predictors as those in Study 1 and Study 2; that is, $X_j = Z_j$. The first column of X_j and Z_j was constructed with ones as the basis coefficients for intercepts. The remaining elements of X_j and Z_j were generated from a MVN distribution ~MVN(0, Σ), where Σ was chosen based on the preset multicollinearity status.

Variables in X_j and Z_j were the candidates for the fixed effects and random effects, respectively. The total number of variables in X_j and Z_j were set to be equal, that is, $p = q = 10$. The five coefficients (β_1 to β_5) and three variances in covariance matrix D were designed not equal to zero. Thus, the five fixed effects (X_1 to X_5) and three random effects (Z_1 to Z_3) were expected to be important. The rest of X_j and Z_j were designed to be nonimportant variables, so the corresponding coefficients and variance components were fixed at zero. The proportions of the important variables were set to be 5/10 for the fixed effect and 3/10 for the random effect.

For the binary outcome, the clustered data were generated using the following logistic generalized linear mixed model (GLMM):

$$\log\left(\frac{p_{ij}}{1-p_{ij}}\right) = x_{ij}\beta + z_{ij}\mu_j + \varepsilon_j, \tag{A1}$$

where p_{ij} denotes the probability of success (i.e., the outcome coded as 1) for the ith observation in the jth cluster, with all other notations defined identically with those in Study 1 and Study 2.

The parameter values of this GLMM were chosen based on prior simulation studies on variable selection (Hui et al., 2015). The true fixed-effects coefficient β was set to β = (−0.1, 1, −1, 1, −1, 0, …, 0), and the true random-effect covariance matrix was consistent with Studies 1 and 2, where:

$$D = \begin{bmatrix} 9 & 4.8 & 0.6 & \dots & 0 \\ 4.8 & 4 & 1 & \dots & 0 \\ 0.6 & 1 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}. \tag{A2}$$

The manipulated factors included within-group size, number of groups, and multicollinearity. We generated 200 replicates per condition in Study 3. The within-group size was designed to be $n = 5$, 10, or 30, representing small to large samples in social sciences (Maas & Hox, 2005; Meinck & Vandenplas, 2012). The group sizes were set at $m = 15$, 30, 50, and 100, reflecting a small to large number of groups (Kreft & de Leeuw, 1998; Krull & MacKinnon, 2001; Maas & Hox, 2005). Under the multicollinearity conditions, we imposed a high correlation of ρ = .9 on either the important variables (x_2, x_3), (x_4, x_5), or the nonimportant variables (x_7, x_8; Kim et al., 2019). The correlations among all other variables were set to ρ = .05. Being consistent with the simulation designs in

Table A1
Average Power Rate for Fixed Effects, Random Effects, and Combining Both Effects in Study 3

	Factor			Fixed effect		Random effect		Combining both effects	
Scenarios	Group size	Within-group size	multicollinearity	RWE	EP	RWE	EP	RWE	EP
$n_j > p$	15	30	No	0.97	.95	0.85	.83	0.83	.79
	15	30	Yes	1	.98	0.9	.66	0.9	.65
	30	30	No	1	1	0.92	.9	0.92	.9
	30	30	Yes	0.99	.99	0.98	.95	0.98	.94
$n_j < p$	100	5	No	0.98	1	0.63	.55	0.61	.55
	100	5	Yes	0.97	1	0.94	.68	0.92	.68
	50	10	No	0.99	1	0.72	.72	0.72	.72
	50	10	Yes	0.99	.99	0.92	.8	0.91	.79

Note. RWE = ranking worth estimation; EP = empirical probability.

(*Appendix continues*)

Studies 1 and 2, we conducted this simulation under two types of scenarios: $n_j > p$ and $n_j < p$, which was consistent with the designs in Studies 1 and 2. The process of applying for the random rPQL was the same as Studies 1 and 2. The results are reported in Table A1.

Results

Overall, random rPQL performed competitively well in selecting important fixed and random effects for a binary outcome in GLMMs. The power rates shown in Table S2 in the online supplemental materials followed the same patterns observed in Study 1 and Study 2, although the power rates were slightly lower. This decrease is consistent to the findings from previous studies in which the regularization approaches exhibit lower power rates in GLMMs with binary than continuous outcomes. Notably, 0.67 was the highest reported power rate among those prior studies. It is much lower than our random rPQL, which is up to 0.83 under similar conditions (group size = 15, within-group size = 30, no multicollinearity), using the ranking worth estimation selection criterion.

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